Lebanese university
Faculty of engineering
Branch III

الجامعة اللبنانية
كليّة الهندسة
الفرع الثّالث

Individual Contribution Report

Smart Climate-Controlled Greenhouse System

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Role: Environmental Monitoring & Ventilation System Engineer

1. Contribution Summary

In the development of the Smart Greenhouse system, my contribution focused mainly on the environmental monitoring subsystem responsible for temperature and humidity measurement and automatic ventilation control.

My work included the integration of the DHT11 temperature and humidity sensor with the Arduino Mega microcontroller, designing the control logic for the ventilation fan, participating in the physical assembly of the greenhouse structure, and assisting in the preparation of the actuator power supply system.

Through this subsystem, I contributed to creating an automatic climate regulation loop capable of detecting temperature variations and activating the cooling fan when necessary.

2. Greenhouse Mechanical Assembly

During the first stage of the project, I collaborated with Razane in assembling the greenhouse base structure.

My responsibilities included:

- Fixing the mechanical base components.
- Preparing mounting locations for electronic elements.
- Ensuring the structure was stable before installing sensors and actuators.

Figure 1: Greenhouse base assembly

3. Environmental Monitoring Subsystem Design

A. DHT11 Sensor Integration

The DHT11 sensor was integrated with the Arduino Mega to continuously monitor the internal greenhouse temperature and humidity.

The sensor provides digital output signals containing:

- Temperature value (°C)
- Relative humidity (%)

The Arduino reads these values periodically and uses them to determine whether ventilation is required.

B. Fan Control Circuit

Since the Arduino cannot directly supply enough current to operate the fan motor, a relay module was used as an intermediate switching element.

The control sequence is:

1. DHT11 measures temperature.
2. Arduino processes the received data.
3. If temperature exceeds the defined threshold:
   a. Arduino sends a HIGH signal to the relay.
   b. Relay connects the external power source to the fan.
   c. Fan starts operating.
4. When temperature returns to no
   a. Relay switches OFF.
   b. Fan stops.

Figure 2: DHT11 sensor and fan components

4. Firmware Development

The Arduino code was developed to create an automatic ventilation algorithm.

The software tasks include:

- Initializing the DHT11 sensor.
- Reading temperature and humidity values.
- Processing sensor data.
- Comparing temperature with the desired limit.
- Sending control commands to the relay.

The control algorithm:

```
Read temperature

If temperature > threshold:
       Activate fan

Else:
       Turn OFF fan
```

```
// ---------------------------------------------------
if (currentTime - fanTimer >= 2000) {
  fanTimer = currentTime;
  float currentTemp = getTemperature();

  if (!isnan(currentTemp)) {
    if (currentTemp > 30) {
      digitalWrite(FAN_RELAY, LOW);
    } else {
      digitalWrite(FAN_RELAY, HIGH);
    }
  }
}
```

Figure 3: DHT11 and fan control code

5. Actuator Power Supply Contribution

I also participated with the team in preparing the external power supply system for the actuators.

The purpose of this system was to provide sufficient current for motors without overloading the Arduino board.

The power system consisted of:

- Two Li-ion 3000 mAh batteries connected in series.
- Step-down voltage regulators.
- Separate supply for actuators.

This configuration ensured stable operation of the fan, pump, and servo motor.

6. System Testing and Validation

After completing the integration of the DHT11 and fan subsystem, testing procedures were performed.

The tests included:

- Checking sensor readings.
- Verifying communication between DHT11 and Arduino.
- Testing relay response.
- Confirming automatic fan activation.
- Checking integration with the complete greenhouse system.

The results showed that the ventilation subsystem responded correctly to temperature variations.

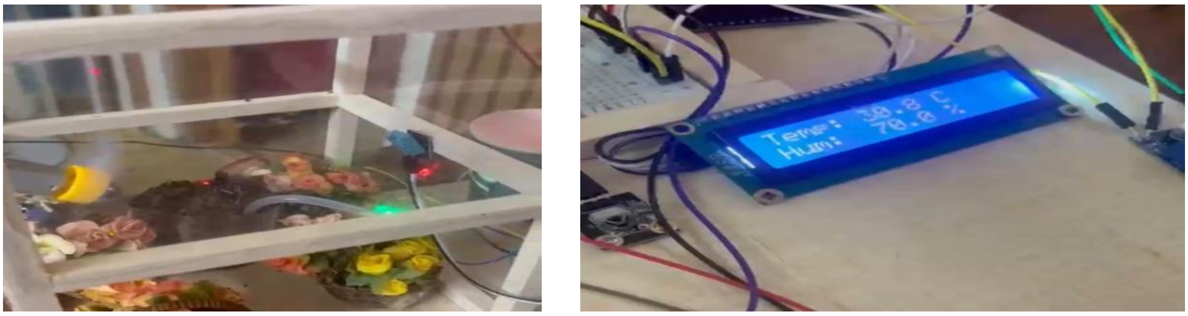

Figure 4: Testing the fan response

7. Personal Reflection

Working on this project improved my understanding of embedded control systems and sensor-based automation.

The main technical challenges were:

- Understanding sensor communication protocols.
- Integrating hardware and software together.
- Managing actuator power requirements.

This project provided practical experience in Arduino programming, electrical connections, relay control, and real-world automation applications.